



Course Code & Title	ETCCCM102(Computational Mathematics II)
Programme & School	B.Tech CSE (School of Engineering and Technology)
Faculty Name	Dr. Vishal
Type of Assignment	Handwritten and Upload
Weightage / Marks	10
Submission Mode	LMS

Assignment 3: Probability and Distributions

Instructions:

- Solve all questions showing detailed steps.
- Use A4 size sheets.
- Submit neatly written solutions.

Questions:

1. If X is a Poisson variate such that

$$P(X = 1) = 2P(X = 2),$$

find:

- (i) $P(X = 0)$
- (ii) Mean
- (iii) Variance

[CO3]

2. With the usual notations, find p for a binomial variate X if

$$n = 6 \quad \text{and} \quad P(X = 4) = P(X = 2).$$

[CO3]

3. In a binomial distribution consisting of 5 independent trials, the probabilities of 1 and 2 successes are 0.4096 and 0.2048 respectively. Find the parameter p of the distribution. [CO3]

4. If X has a Poisson distribution such that

$$P(X = 1) = P(X = 2),$$

find $P(X = 4)$. [CO3]

5. If X is a random variable following a binomial distribution with mean 2.4 and variance 1.44, find:

- (i) $P(X \geq 5)$
(ii) $P(1 < X \leq 4)$

[CO3]

6. If X and Y are independent Poisson variates such that

$$P(X = 1) = P(X = 2) \quad \text{and} \quad P(Y = 2) = P(Y = 3),$$

find the variance of $X - 2Y$. [CO3]

7. If X is a Poisson variate such that

$$P(X = 2) = 9P(X = 4) + 90P(X = 6),$$

find:

- (i) λ
(ii) Mean of X

[CO3]

8. Ten coins are thrown simultaneously. Find the probability of getting at least seven heads. [CO3]

9. Comment on the following statement:

"The mean of a binomial distribution is 3 and variance is 4." [CO3]

10. The mean and variance of a binomial distribution are 4 and $\frac{4}{3}$ respectively. Find $P(X \geq 1)$. [CO3]

11. Suppose that X has a Poisson distribution. If

$$P(X = 2) = \frac{2}{3}P(X = 1),$$

evaluate $P(X = 0)$.

[CO3]